Date 10/9/26
Page 1

Atom

The smallest unit of any matter which cannot be divided further easily or by easy process.

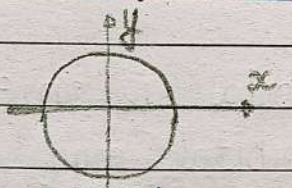
Orbit :-

An orbit is a circular imaginary on which the electron (e^-) revolve around its nucleus.

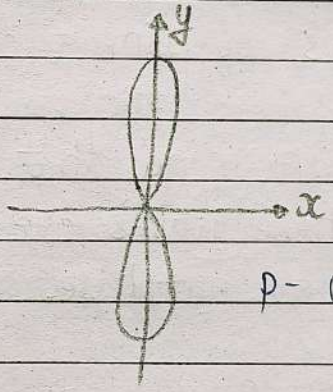
Orbital :-

Orbital is a 3D space around nucleus where the probability of finding e^- is maximum.

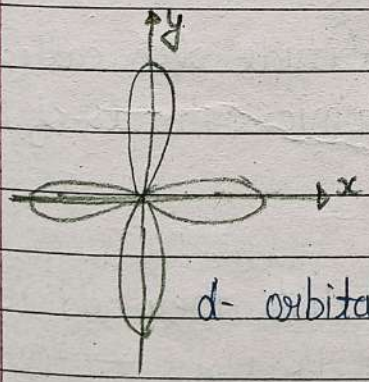
Shape of Orbitals:



s- orbital



p- orbital



d- orbital

f- orbital ⇒ complex

Molecular Orbital Theory:

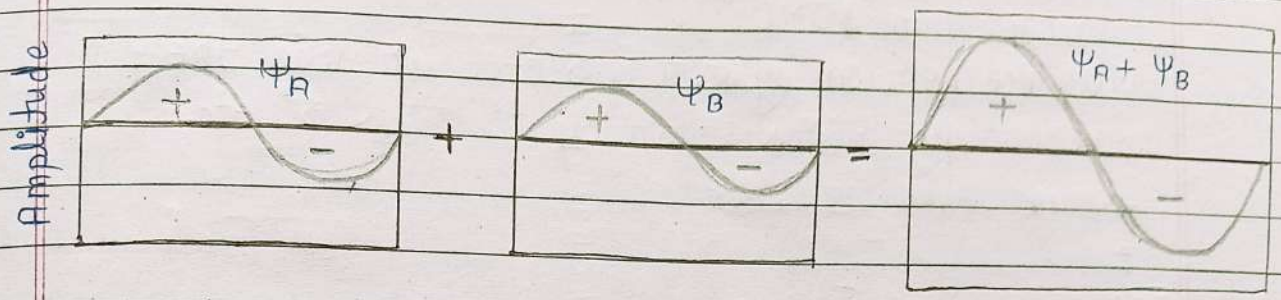
This theory was given by F. Hund & R.S Mulliken.

- According to this theory, when the atomic orbitals are combine together to form molecular orbital. The individual identity of atomic orbital is lost.
- When two atomic orbitals combine together they form two new orbitals called Bonding Molecular Orbitals (BMO) and Anti-Bonding Molecular Orbital (ABMO).
- The formation of Molecular orbital can explained on the basis of linear combination of atomic orbitals.
- Keeping in mind that an atomic orbital is an electron wave, these waves are two atomic orbital may be in phase or out phase.
- Suppose Ψ_A & Ψ_B represent the amplitude also called wave function of the e⁻ wave of atomic orbitals of atom A and B respectively.

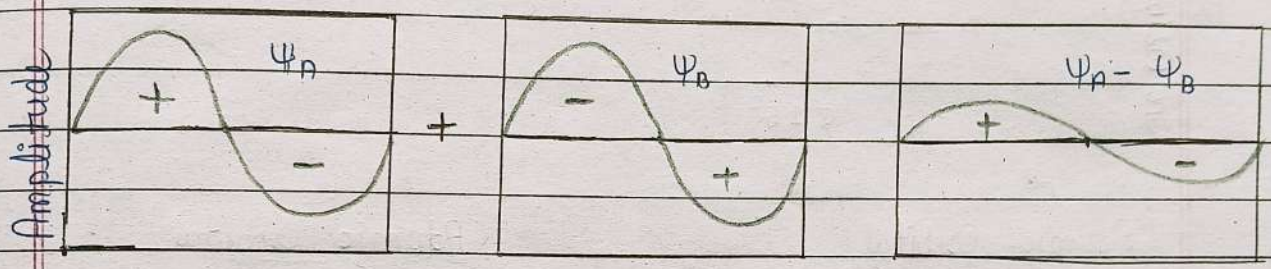
Case-1 When two waves are in-phase so that they add up and amplitude of new wave is $\phi = \Psi_A + \Psi_B$ —— (1)

Case-2 When two waves are out of phase, the waves are subtracted from each other $\phi' = \Psi_A - \Psi_B$ —— (11)

In Phase:



Out Phase:



The probability is (density of e^- cloud) given by the square of the amplitude.

$$\boxed{(\phi^2 = \Psi_A + \Psi_B)^2}$$

$$\phi^2 = \Psi_A^2 + \Psi_B^2 + 2\Psi_A.\Psi_B \quad \text{——} \ (3)$$

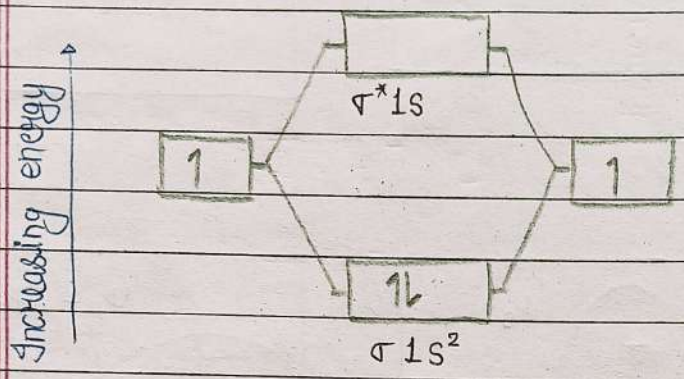
$$\phi^2 = \Psi_A^2 + \Psi_B^2 - 2\Psi_A\Psi_B \quad \text{——} \ (4)$$

i.e: $\phi^2 > \Psi_A^2 + \Psi_B^2$ where as $[(\Psi_A + \Psi_B)^2 = (\Psi_A^2 + \Psi_B^2 + 2\Psi_A\Psi_B)]$

$$\phi^2 < \Psi_A^2 + \Psi_B^2$$

Thus, the combination of two atomic orbital, two new orbital are formed by the subtractive effect. The molecular orbital formed by additive effect is called Bonding molecular Orbital and molecular orbital formed by the subtractive effect called Antibonding molecular Orbital.

ABMO has more energy, hence they have ~~greater~~ lesser stability than BMO.
BMO are shown by σ, π & δ etc. whereas the corresponding ABMO shown by σ^*, π^* & δ^* etc.



A
(Atomic Orbital)

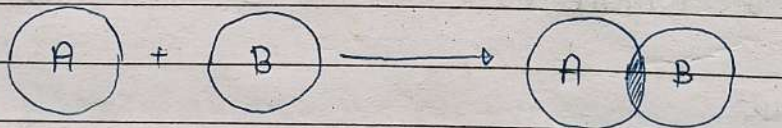
B
(Atomic Orbital)

Pictorial representation of bonding AB of H_2 molecule:

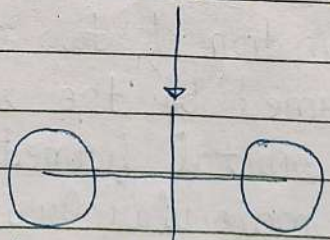


S-orbital S-orbital overlapping by addition Result: BMO

Example of ABMO:



Overlapping by subtraction



Resultant antibonding molecular orbital.

Application of Molecular Orbital Theory for Diatomic molecule:

Diatomic molecule are of two types:

a) Homonuclear diatomic molecule
b) Heteronuclear diatomic molecule

[a] Homonuclear diatomic molecule:

When both the atoms are same, called Homonuclear diatomic molecule.

Ex: Li_2, Be_2, B_2, C_2, N_2, O_2, F_2, Ne_2, H_2 etc.

- Molecular orbital energy order:

For B_2, C_2, N_2:

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

For O_2, F_2:

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x = \pi 2p_y < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

[b] Heteronuclear diatomic molecule:

When both the atoms are different, called Heteronuclear diatomic molecule.

Ex: CO, NO, HF, HCl

- Molecular Orbital energy order:

$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x = \pi 2p_y < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$

Example : 1

H_2 molecule → 2H atom

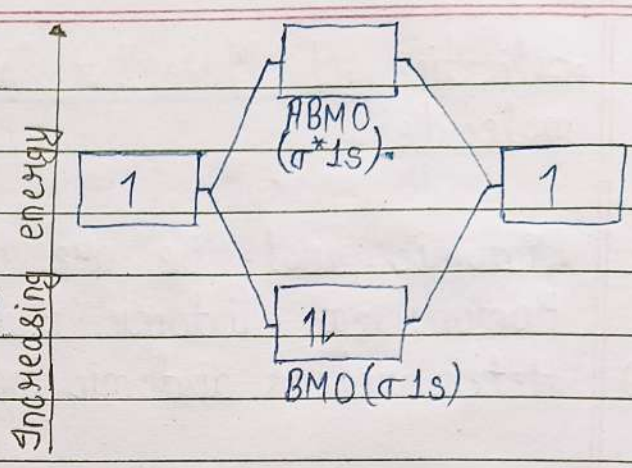
No. of e^- in H_2 atom = 2

Energy level = $\sigma 1s^2 < \sigma^* 1s^0$

Bond Order = $\frac{1}{2}[N_b - N_a]$

$= \frac{1}{2}[2-0]$

B.O = 1



Molecular orbital diagram of H_2

Example - 2

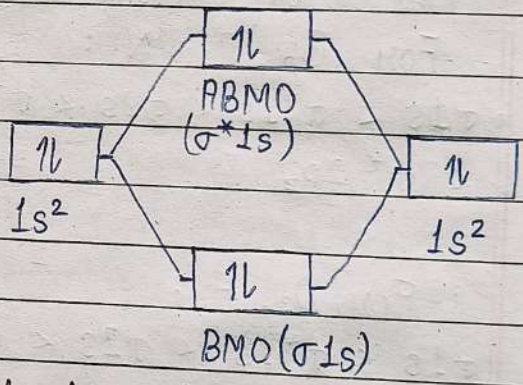
He_2 molecule → 2He atom

No. of e^- in He_2 molecule = 4

Energy level = $\sigma 1s^2 < \sigma^* 1s^2$

B.O = $\frac{1}{2}[2-2] = 0$

Diamagnetic in nature



Molecular orbital diagram of He_2

Example = 3

B_2 molecule → 2B atom

Energy level → $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^1 = \pi 2p_y^1$

B.O = $\frac{1}{2}[6-4]$

B.O = 1

Example - 4

C_2 molecule = 2C atom

Energy level = $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2$

$B.O = \frac{1}{2}[8-4]$

$B.O = 2$

Example - 5

Be_2 molecule = 2Be atom

Energy level = $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2$

$B.O. = \frac{1}{2}[4-4]$

$B.O = 0$

For Be_2^+

Energy level = $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^1$

$B.O = \frac{1}{2}[4-3]$

$= 0.5$

Example - 6

N_2 molecule = 2N atom

Energy level = $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \pi 2p_x^2 = \pi 2p_y^2 < \sigma 2p_z^2 < \pi^* 2p_x = \pi^* 2p_y$

$B.O = \frac{1}{2}[10-4]$

$= 3$

For N_2^+ = 2.5

N_2^- = 2.5

N_2^{2-} = 1

Example-7

O_2 molecule = 20 atom

Energy level = $\sigma 1s^2 < \sigma^* 1s^2 < \sigma 2s^2 < \sigma^* 2s^2 < \sigma 2p_z^2 < \pi 2p_x^2 = \pi 2p_y^2 < \pi^* 2p_x^1 = \pi^* 2p_y^1 < \sigma^* 2p_z^0$

$$B.O = \frac{1}{2}[10-6]$$

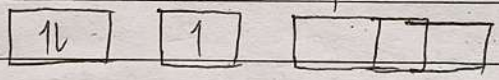
$$= 2$$

Band Theory based on Molecular Orbital Theory:

MOT may be extended to explain bonding in metals and known as Band theory.

The electronic structure of Li atom is-

Li_3 = $1s^2$ $2s^1$ $2p^0$

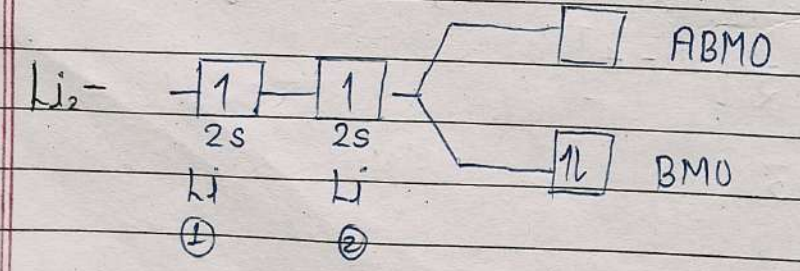


Consider the promotion of Li molecule:

There are $6e^-$ arranged in a molecular orbitals are.

$Li_2 = \sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^0$

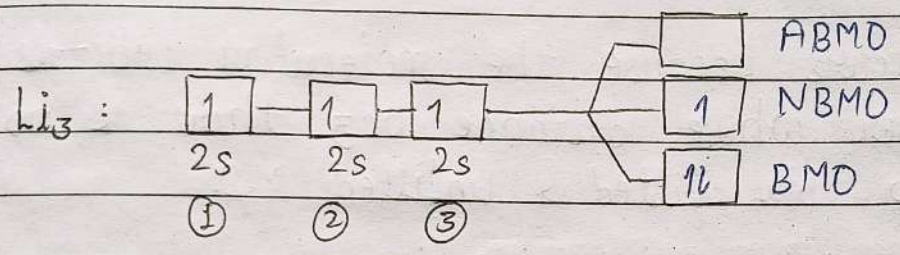
In this bonding occurs because of the $\sigma 2s$ bonding molecular orbital is full and anti-bonding molecular orbital is empty.



In the formation of Li_3 molecule:

3 - 2s atomic orbital combine to form three molecular orbital like one bonding one non-bonding & one antibonding.

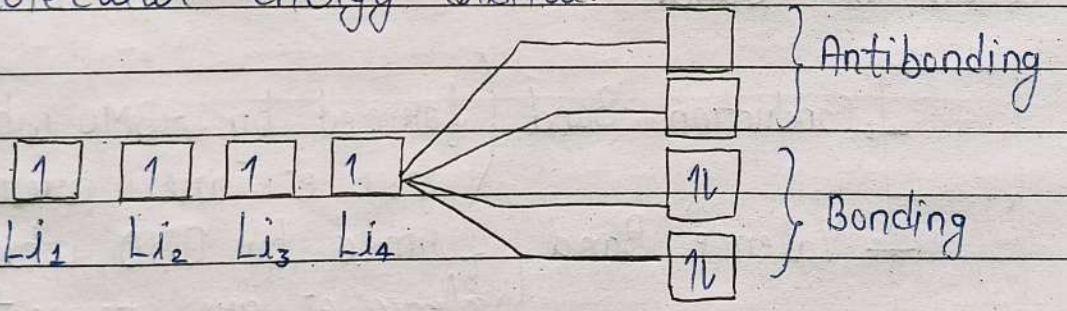
The energy of non-bonding orbital is b/w the bonding & antibonding orbital as shown below:



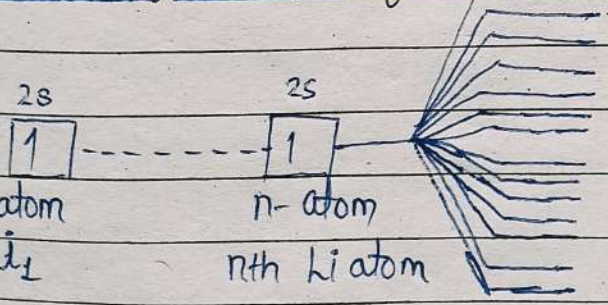
In the formation of Li_4 molecule:

The four 2s atomic orbital forms four molecular orbital : two bonding & two antibonding

The four valence e^- occupy the two lowest molecular energy orbital.



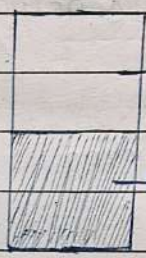
In the formation of Li_n molecule:



Very closely spaced MO levels constituting a band. There are in energy levels of n electrons so that band is only half filled.

Development of molecular orbital into Band in metals:

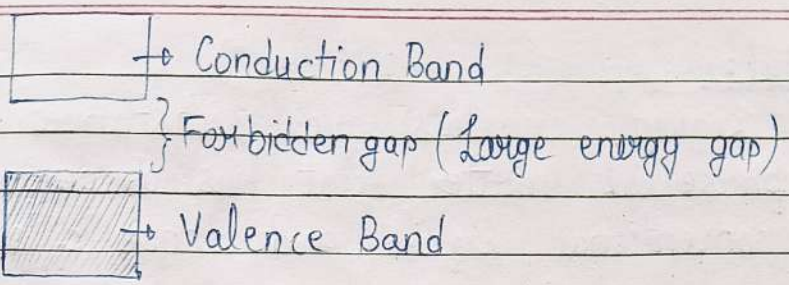
- On the basis of band theory, materials can be classified; If one end of metal piece is heated, electron gain energy move to an unoccupied M.O. where they travel rapidly in metal and metals become hot.
- Conduction occurs because the molecular orbitals extended over whole distance bcz there is no energy gap b/w filled & unfilled M.O.
- The band theory can also explain the classification of materials into conductor, insulator & semiconductor.
- This arises due to the difference in the gap b/w filled & partially filled valence bond & unfilled conduction band.



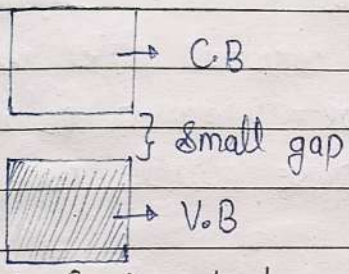
Conduction Band (formed by ABMO where free e⁻ are present)

Valence Band (formed by BMO where valence e⁻ are present)

(a) Conductor



(b) Insulator



(c) Semi conductor

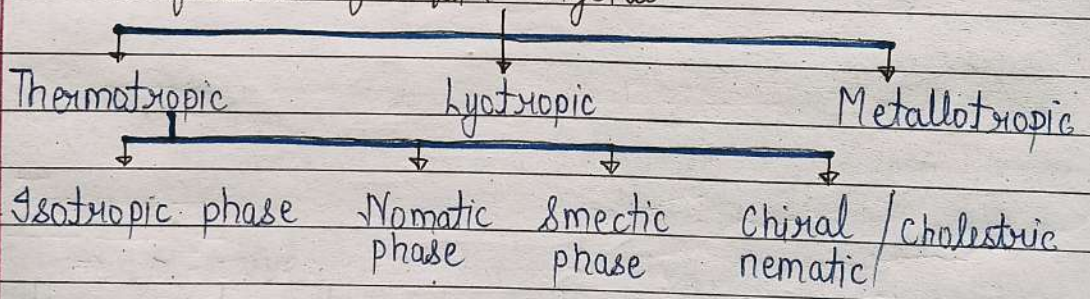
Liquid crystal:

Liquid crystals are substances that exhibit a phase of matter that has properties b/w those of a conventional liquid and those of a solid crystal.

- For instance, a liquid crystal may be flow like a liquid, but have the molecules in the liquid arrangement/ orientation in a crystal like way.
- When viewed under a microscope using a polarised light source, different crystal phase will appear to have distinct texture.

Ex: Cholestrol ester, phenyl benzoate, surfactants paraffins glycolipids, cellulose derivatives, solution of soap and various detergent & tobacco mosaic virus.

Classification of Liquid Crystal:



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Point Defect (Atomic Imperfection):

- Any deviation from perfectly ~~arrange~~ ordered arrangement of atoms in crystal is called Imperfection crystal or called defect.

Point Defect: When deviation exists from the regular (or periodic) arrangement around an atom or group of atom in an crystalline substance is called Point defect

Point Defect

(a) Stoichiometric. Ratio of cation & anion remain same (1:1)
- Schottley Defect
- Frenkel Defect

(b) Non-stoichiometric Ratio of cation & anion is different
- Metal excess
- Metal deficient

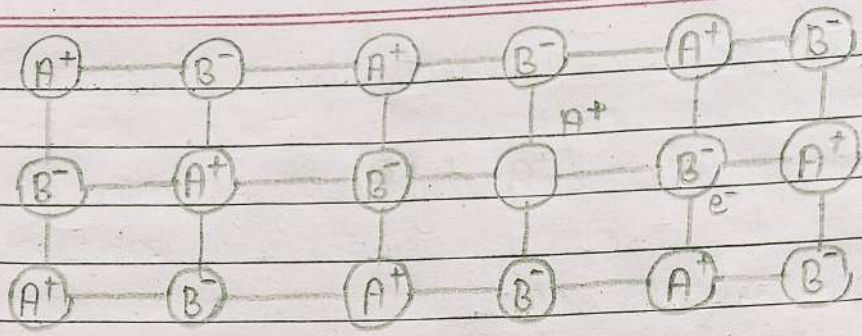
(c) Impurity When foreign atom are present at lattice sites (dopped)
- n-type
- p-type

Metal Excess:

Ex: When ZnO is heated it loses oxygen & turn yellow due to the following reaction

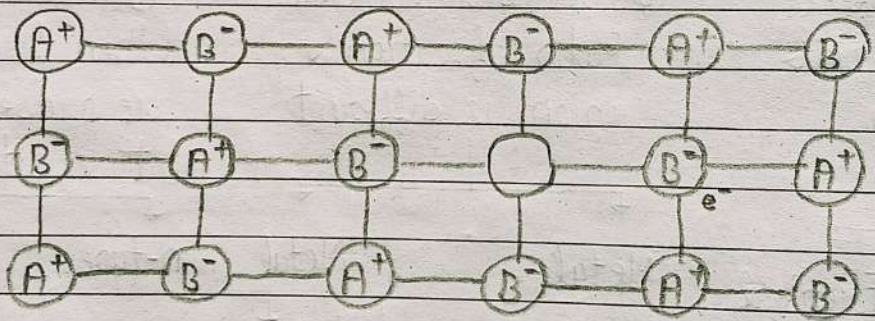
$$ZnO \xrightarrow{\Delta} Zn^{+2} + \frac{1}{2}O_2 + 2e^-$$

Thus excess of Zn^{+2} ion thus form get appeared into the vacant interestial site and e^- in the neighbouring interestial site.



Metal Deficient:

This defect occur when the metal show variable O.S, i.e transition metal. The defect usually occur due to the missing of cation from its lattice site and the presence of cation having higher charge. (+2 instead of +1) in the lattice site

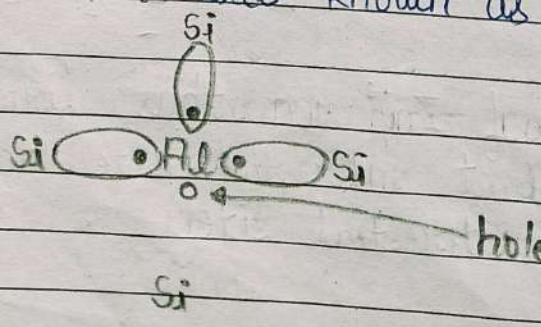


Impurity

⇒ p - type:

In p- type defect, the 3rd group ~~elemet~~ element is added as a doping element trivalent impurity like Al, Ga, In added in this defect.

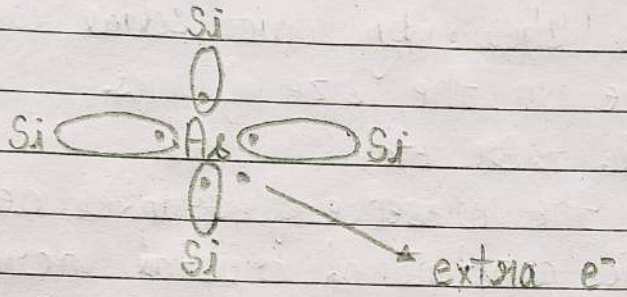
- It provides extra holes known as a acceptor atom.



⇒ n- type:

5th group element is doping element.

- Pentavalent like N, P, As, Sb, Bi.
- It provides extra e^- and act as a donor atom.

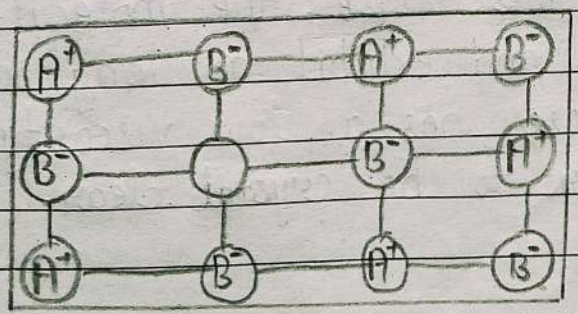


Point Defect (Zero Dimension):

Point defect are the irregularities or deviation from ideal arrangement around a point or an atom in a crystalline substance.
These are of two types:

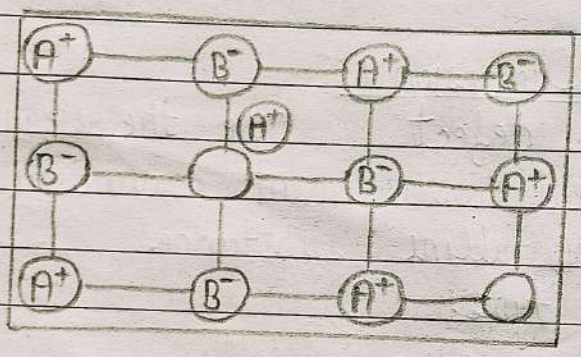
Schotty Defect:

- It is a type of vacancy defect in ionic crystals. The no. of missing cation and anions is equal to maintain electrical neutrality of ionic solid.
- In this defect the density of substance decreases.
- This defect is shown by ionic substances which have almost same size of cation and anion
- Ex: NaCl, KCl, $AgCl_2$, CsCl, AgBr etc.



Frenkel Defect:

- In this defect, the smaller ion, usually cation get delocated from its normal site to an interstial position.
- This defect is shown by ionic solids which have large difference in the size of ions.
- Ex: ZnS, AgCl, AgBr etc.
- ~~It is~~ Due to the presence of frankel defect, dielectric constant of crystal increases.



Liquid Crystal continues:

[1] Thermotropic: The transitions between the liquid crystals are given by the change in temperature and hence these are called Thermotropic.

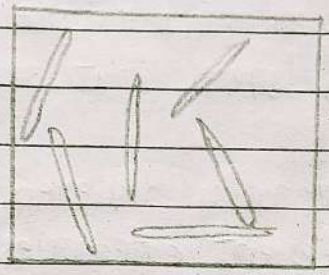
- Thermotropic consists of organic molecule.
- Thermotropic are mostly used for technical applicators.
- Thermotropic liquid crystals are classified as-

(i) Isotropic Phase:

In this phase, the molecules are randomly aligned and exhibit no long range order.

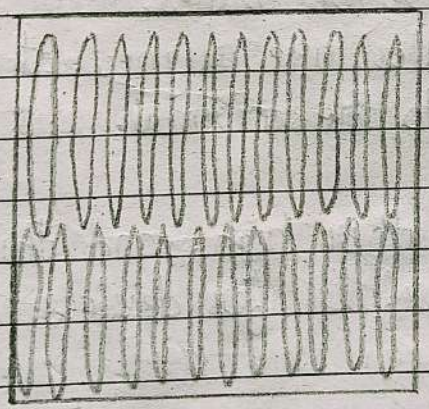
→ The isotropic phase has a low viscosity and will often appear to be crystal clear.

→ For all practical purposes, the isotropic phase microscopically appears to be like any other isotropic liquid such as water.



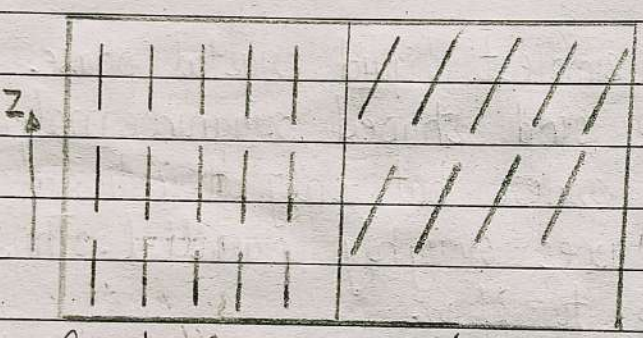
(ii) Nematic phase:

- The simplest liquid crystal phase.
- In this phase, the rod shaped organic molecule have no positional order but only directional order, with their long axes roughly parallel. Thus their molecules are free to flow.
- These have optical properties and extremely useful in Liquid crystal displays (LCD).
- The digital water you used to wear back in 80's functioned using nematic liquid crystals.



[iii] Smectic phase:

- 'Smecticus' means cleaning or having soap like properties.
- This is found at lower temperature than the nematic & form well defined layers that can slide over one another like soap.
- Thus, these are positionally ordered along one direction.
- There are several different categories to describe smetics.



- The two best known are -

Smectic A - in which the molecules are aligned ⊥ʳ to the layer planes.

Smectic C - molecules are aligned at some arbitrary angle to the normal.

- The viscosity is rather high, and this phase is generally not useful for devices.

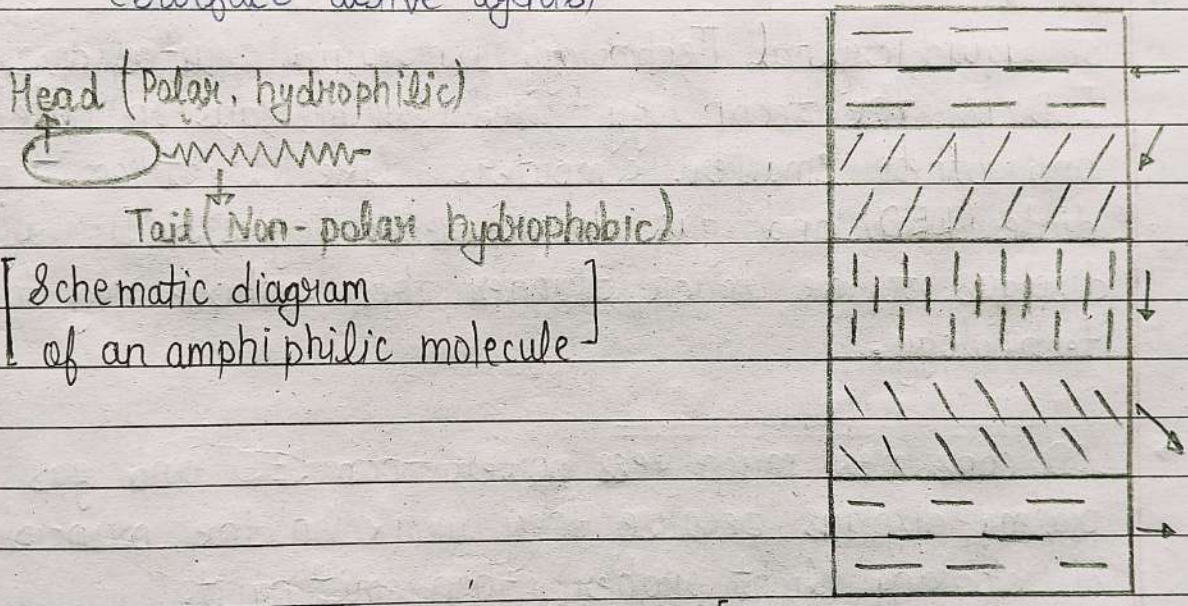
[iv] Chiral Nematic / Cholesteric :-

- It was first observed in cholestrol derivatives.
- In this phase, the molecules twist slightly from one layer to the next, resulting in a spiral formation.

[2] Lyotropic Liquid Crystal:

- Lyotropic Liquid Crystal phases are formed by amphiphilic (both hydrophilic and hydrophobic) molecules.
- Lyotropic Liquid Crystalline phases are abundant in living systems.

Ex: Biological membranes, cell membranes, surfactant (surface active agents)



[Schematic diagram of an amphiphilic molecule]

[Schematic diagram of chiral nematic molecule]

→ Lyotropic Liquid Crystal:

The phase transition may also depend on concentration of the mesogen in the solvent, and these liquid crystals are called Lyotropic.

[3] Metallotropic -

- consist of both organic and inorganic molecule.
- Their liquid - crystal transition depends on temperature concentration & the ratio of inorganic - organic composition.

Ex: $ZnCl_2$

Application of LC :

[1] Liquid Crystal Displays (LCD):

→ The most common application of liquid crystal Technology is liquid crystal displays (LCDs). LCD is the technology used for displays in notebook and other smaller computers. Like Light emitting diode (LED) and gas plasma technologies, LCDs allow displays to be much thinner than cathode ray tube (CRT) technology.

→ LCD consumes much less power than LED and gas plasma displays because they work on the principle of blocking light rather than emitting it.

[2]. Liquid Crystal Thermometers -

Chiral nematic liquid crystals used as crude liquid crystal thermometers.

[3]. Optical Imaging:

An application of liquid crystals is optical imaging and recording which is the most promising areas of liquid crystal research.

[4] In Industry:

The liquid crystal sheets used for hot spots, map heat flow, measure stress distribution patterns and so on.

[5] LC lasers used in lasing medium.

[6] Liquid crystals are used as solvents for studying the structure of anisotropic molecule spectroscopy.

[7] Liquid crystals are used for detecting tumours in the body by using a method called thermography.

[8] Liquid crystals are used as commercial Lubricants.

[9] Liquid crystal are used in display devices such as LCDs (Liquid Crystal Displays).

[10] Liquid crystal are most suited to biological functions. As the colour of liquid crystals depend sensitively on temperature, these are therefore, used in measuring skin temperature.

Nano Material and Analytical Instrumental Techniques:

- The 'nano' term comes from greek word DWARF.
- $1nm = 10^{-3}\mu m = 10^{-6}mm = 10^{-9}m$
- Nanomaterial exhibit completely different properties compound bulk material.
- As dimensions of material are reduced, the optical electronic, magnetic, mechanical and chemical properties are changed drastically

Ex: Nano gold particles are excellent catalysts. while bulk gold is inert.
Nano silver particles exhibit anti-bacterial properties while bulk silver does not.

- The unique behaviour of nano materials can be accounted due to (i) large surface area & (ii) quantum effect.
- There are several ways to classify nano-materials. The two extreme nano materials are:

[A] Based on no. of dimensions:

(i) Zero Dimension:

- The length, width & height are confined at single point i.e., quantum dots or nano sphere, nano clusters and textile from exposure to UV rays.
- Silver nano-particles are being used to cleanup CCl_4, pollution in ground water.
  Iron oxides nano particles are being used arsenic from water wells.
- Silver nano-particles are used to form the conducting lines needed in circuit boards.

Carbon Nanotubes (CNTs)

- CNTs are allotropes of carbon with a cylindrical nanostructure.
- A single sheet of graphite is called graphene. CNTs are formed by rolling a graphene sheet into a cylinder.
- The CNTs are elongated forms of fullerenes. sheet rolled into the shape of a cylinder.
- CNTs are shaped materials made up of carbon, having a diameter ranging from 1 to 15 nm and a very large/constant length.

- CNTs can naturally coil to form ropes, which are held together by Vander Waals forces, specifically by π-π stacking.
- Nanotubes are categorised as:
  - → ~~Simple~~ - Single - walled nanotubes (SWTs)
  - → Multi - walled nanotubes (MWTs)

Properties of CNTs:

- → They exhibit extraordinary strength due to C-C bonds.
- → They have higher elasticity; the tube can recover from severe structural distortion.
- → The low density of nanotubes makes them extremely lightweight materials.
- → Nanotubes have high resistance to chemical attack & oxidation.
- → The thermal conductivity of nanotubes is about two times higher than that of diamond.
- → Their electrical conductivity is about 100 times higher than that of copper.

[ii] One dimension:

- It has only one parameter either length or breadth or height.

Eg.: Quantum wires, nano wire, nano rods

[iii] Two dimension:

- It has only length and width.

Eg: Nano tubes and nano films.

[iv] Three dimension:
It has all parameters length, width & height.
Ex: Nano particles.

(B) Based on Composition:
(a) Carbon based materials. Eg.: Composed of carbon, tubes etc.
(b) Metal based materials. Eg.: Nanogold, nanosilver & TiO_2 etc.
(c) Dendrimers (Repeatively branched molecule)
(d) Composites (Combination of nanoparticles with other nano-particles)

Application of Nano-material:

Nano-technology
- Medicine and health care
  - Cancer treatment
  - Tissue Eng^{n}
  - Finding of DNA sequence
  - Anti-oxidant activities
  - Labelling of biological molecule
  - Nano cluster foods
  - Eye surgery
  - Nano filteration
  - Catalysis
  - Sunscreen lotion
- Power sector
  - Nano semi-conductor
  - Recycling the batteries
  - Nano LEDs
- Electronics
  - High speed micro processor
  - Enhance memory devices

- Information technology → Quantum computers.
- Other applications
  - → Aerospace
  - → Construction
  - → Packaging
  - → Nano-filteration
  - → Catalytic converters
  - → Automobile components
  - → Nanopoly ~~UV ethane tyre~~ urethane tyres

Application of carbon nano tubes:

- Due to high electrical conductivity, nanotubes are used in electronic devices such transistors.
- They are used as a hydrogen storage system in fuel cells.
- By using nanotube filters, it is possible to filter out bacteria and viruses from water.
- They are used as a composite for batteries and television monitors.
- They are particularly used in a electronic field to create non-volatile memory.

Nano-particles:

- Nano particles are particles b/w 1-100 nm in size.
- Nano particles are different from bulk material & isolated molecules because of their unique optical, electronic & chemical properties.
- Reduction in size is responsible for the enhanced properties
- The unexpected properties of nanomaterial are due to the large surface area of the particle.
  Ex: Gold nanoparticles appears deep red to black in solution and magnetic.
- Observation of solar radiation is much higher in material composed of N.P. than it is in thin film.
- Magnetic nanoparticles are used in arrange of application. like imaging, bioprocessing, refrigeration & high storage density magnetic memory media.
- Suspension of N.P are possible since the interaction of the particles surface with the solvent is strong.
- The high surface area to volume ratio of N.P provides a tremends driving forces for diffusion.

Applications:

- Nanomaterials used to deliver drugs to tumors.
- Polymer coated iron oxide nanoparticles used to break up clusters of bacteria, possibly allowing more effective treatment of chronic bacterial infections.
- Carbon nanoparticles are used in medical applications such as treat leukemia.

- Zinc oxide nanoparticles can be dispersed in industrial coatings to protect wood, plastic and textiles from exposure to UV rays.
- Silver nanoparticles in fabric are used to kill bacteria, making clothing odor-resistant.
- Iron nanoparticles are being used to clean up carbon tetrachloride pollution in ground water. Iron oxide nanoparticles are being used to clean arsenic from water wells.
- Silver nanoparticle ink was used to form the conductive lines needed in circuit boards.

⚠ **Carbon nanotubes:** (Repeated)

- Carbon nanotubes (CNTs) are allotropes of carbon with a cylindrical nanostructure.

Nanoclusters:

- Nanoclusters are intermediate state b/w molecules and bulk materials.
- They are aggregate of atoms or molecules.
- A nanocluster is a grouping of a no. of nanoparticles in a narrow size distribution having atleast one dimension between 1 and 10 nanometers.
- Forces like metallic, ionic, hydrogen bond or weak Van der Waal's force of attraction binds these atoms or molecules.
- The magic no. is the no. of atoms present in the cluster with critical sizes and high stability. The cluster can have 2 to 2000 atoms.

- Cluster with special magic no. are predicted to be more stable and to show special features compared to other.
- Cluster of different sizes will have different optical properties, electronic structures and different energy level separations.
- Nanoclusters are of technological interest in numerous areas of applied science such as materials science, catalysis.

Applications:

- Because of changes in electronic and magnetic properties, their applications range from nonlinear optical switching and catalysis to high-density information storage.
- Magnetic nanoclusters such as Fe, Co, and Ni are also of immense technological importance due to their preferred orientations.
- Nanoclusters are used in microelectronics, tele-communication, sensors, transducers, biological markers, switches, electroluminescent displays, chemical reactors, catalysts etc.
- Nanoclusters are also used in biotechnology and pharmacology.

Synthesis of Nanomaterials:

Nanomaterials deal with very fine structures: a ~~nanomaterial~~ nanometer is a billionth of a meter. The two mainstream approaches to synthesis nano-materials are top-down & bottom-up.

→ **Top down approach:**

Nanomaterials have been prepared by the breakdown of materials using techniques developed by solid state physicists. For example, a large block of silicon wafer can be reduced to smaller components by cutting, etching, & slicing them down to a desired size or shape. This is known as the top-down approach.

→ **Bottom up Approach:**

The bottom up approach, starts with the individual building units that are the objects of the nanometer scale, such as atoms, molecules, polymers and colloids. By assembling these building units with the needed controllability, the desired nanostructured systems can be obtained. Nanomaterials synthesised from the bottom-up method have novel physico-chemical properties that differ from the bulk material.



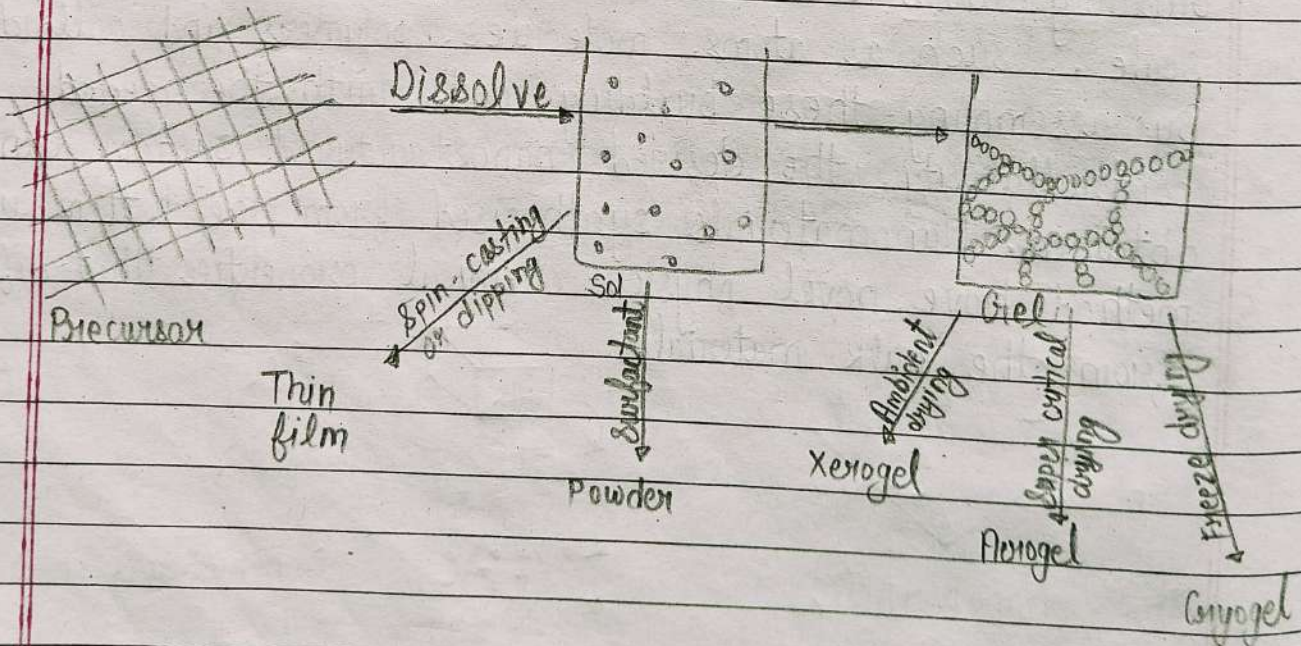
Top-Down Method

- High energy Ball Milling
- Lithography
- Gas Condensation
- Severe plastic deformation

Bottom-up Method

- Physical Vapour Deposition
- Chemical Vapor Deposition
- Plasma Processes
- Sol-gel processing
- Soft Lithography
- Self assembly

1> **Sol-Gel Method ⇒**



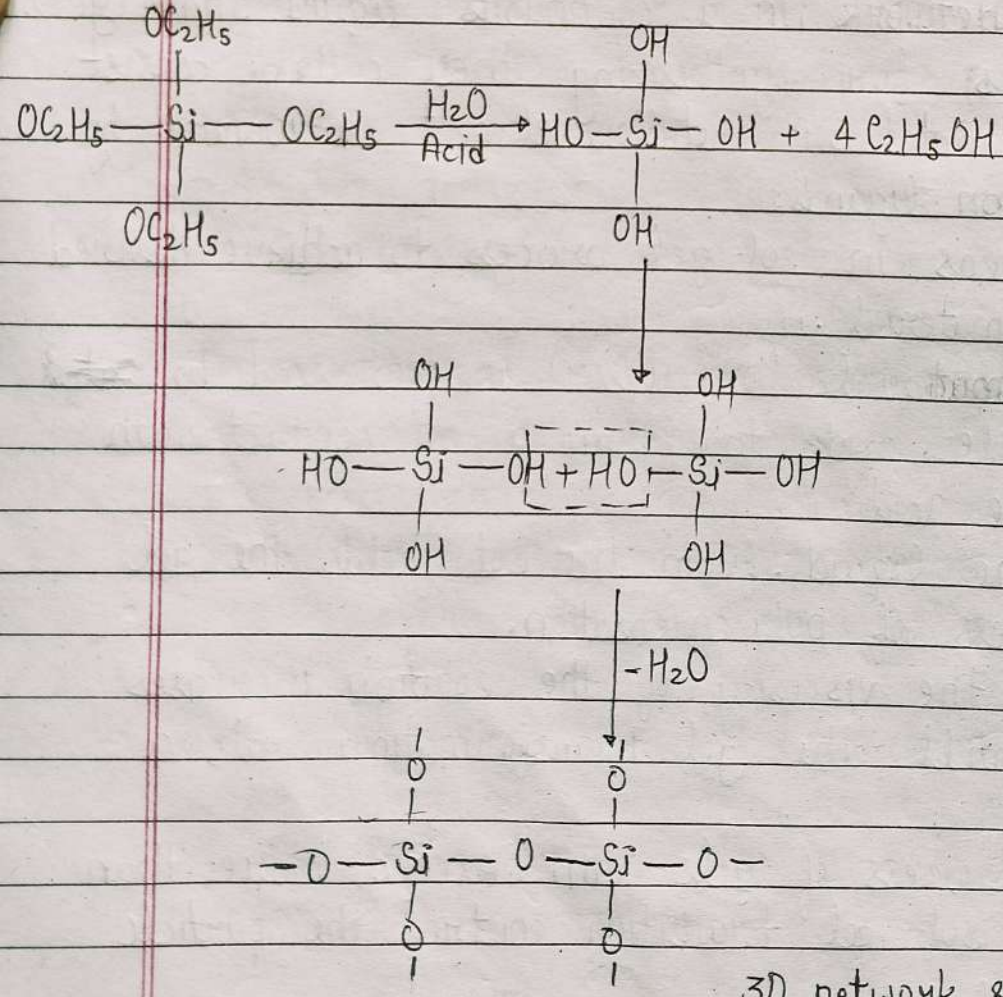
- $MOR + H_2O \longrightarrow MOH + ROH$ (Hydrolysis)
  (Metal alkoxide) (Metal hydroxide)
- $MOH + ROM \longrightarrow M-O-M + ROH$ (Condensation)
- $MOH + MOH \longrightarrow M-O-M + H_2O$

M → metal R → alkane.

- Sol-gel method of synthesising nanomaterial is very popular amongst chemist and is widely used to prepare oxide material.
- The sol-gel process involve the evolution of inorganic network through the formation of a colloidal suspension (sol) and gelation of the sol to form a network in a continuous liquid phase (gel).
- The precursors for synthesizing these colloide consist usually of a metal or metalloid element surrounded by various reaction ligands.
- The various steps in sol-gel process to achieve desired network (nanomaterial).
- The starting materials (precursor) is processed to form dispersible oxide to form a sol contact with water or dilute acid.
- Removal of the liquid from the sol yields the gel by the process of polycondensation.
- In this step, the viscosity of the solution increases continuous until the gel transformation into a solid mass.
- This aging process of gels can exceed more than 7 days and sol-gel transition control the particle size & shape.

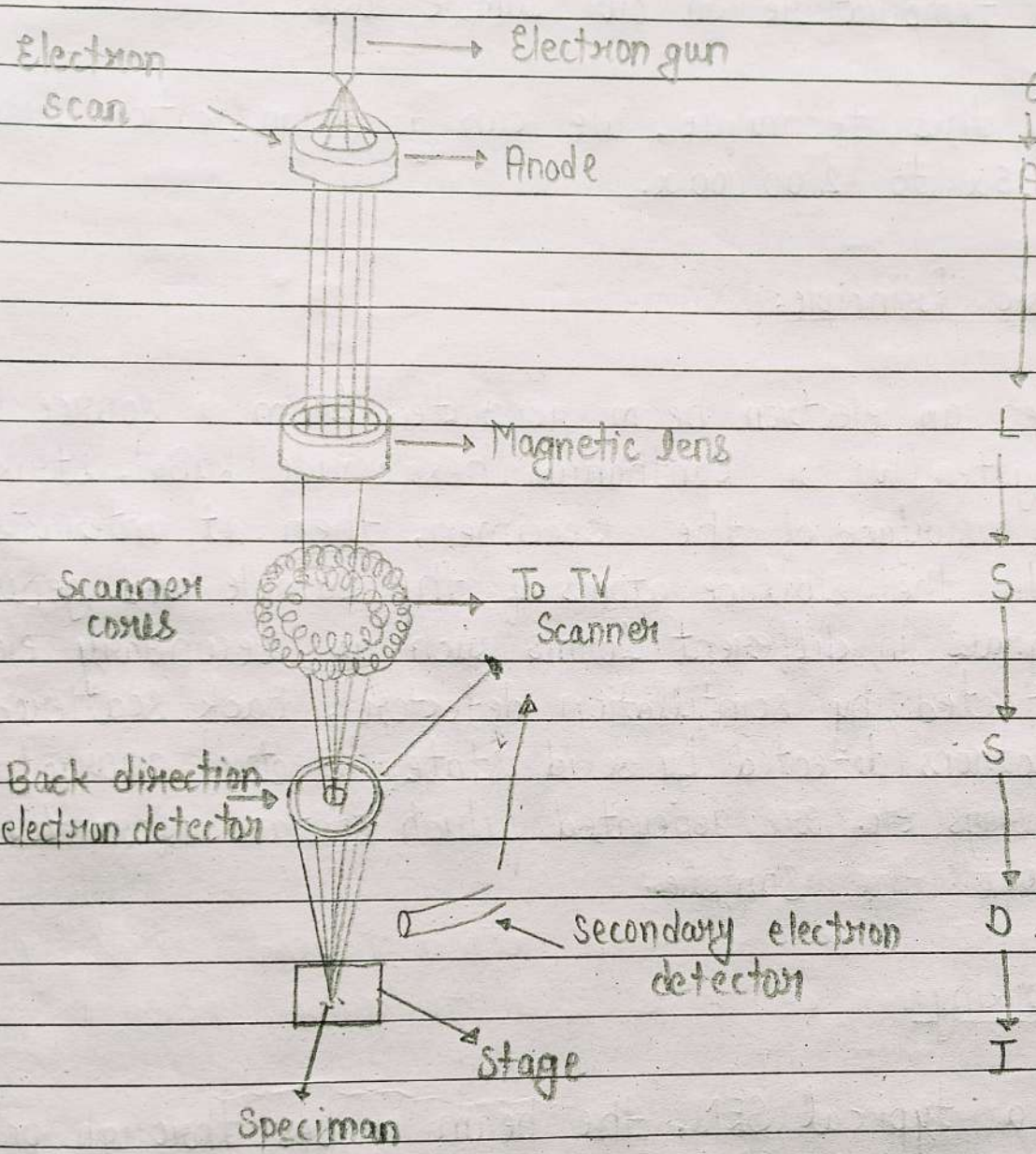
- Drying removal of solvent such as alcohol, water of the gel is critical process is based on isolation type the monolith name will be
- By calcination process of gel, we can get material like dense ceramic (oxide materials) which are generally synthesize by fusion which require very high temperature
- The reaction involved in the sol-gel chemistry based on the hydrolysis and condensation of metal alkoxide can be described as follows:

Eg.: Sol-gel process (i.e. hydrolysis & condensation) of precursors such as
$Si(OC_2H_5)_4$ → Tetraethoxy silane
Tetra ethyl orthosilicate



3D network structure

~~Scanning~~ Scanning Electron Microscope:



- Conventional light microscope uses a series of glass lenses to bend light waves and create a magnified image, while the scanning electron microscope (SEM) creates the magnified images by using electrons instead of light waves.
- By using SEM, we can study about topography, crystallography and chemistry of a sample etc.

- Even we can perform some-in-situ reactions & effect of temperature on our sample also.
- By this technique, we can go magnification from 15x to 2,00,000 x.

Basic Principle:

- When an electron beam generated from a source like Tungsten (W) or Lanthanum hexaboride (LaB_6), strikes the surface of the specimen, then it interacts with the surface atoms of the sample and generate signals in different forms such as secondary electrons (detected by scintillation detector), back scattered electrons (detected by solid state detector), characteristic X-rays etc. are generated which contain information about the sample.

Working:

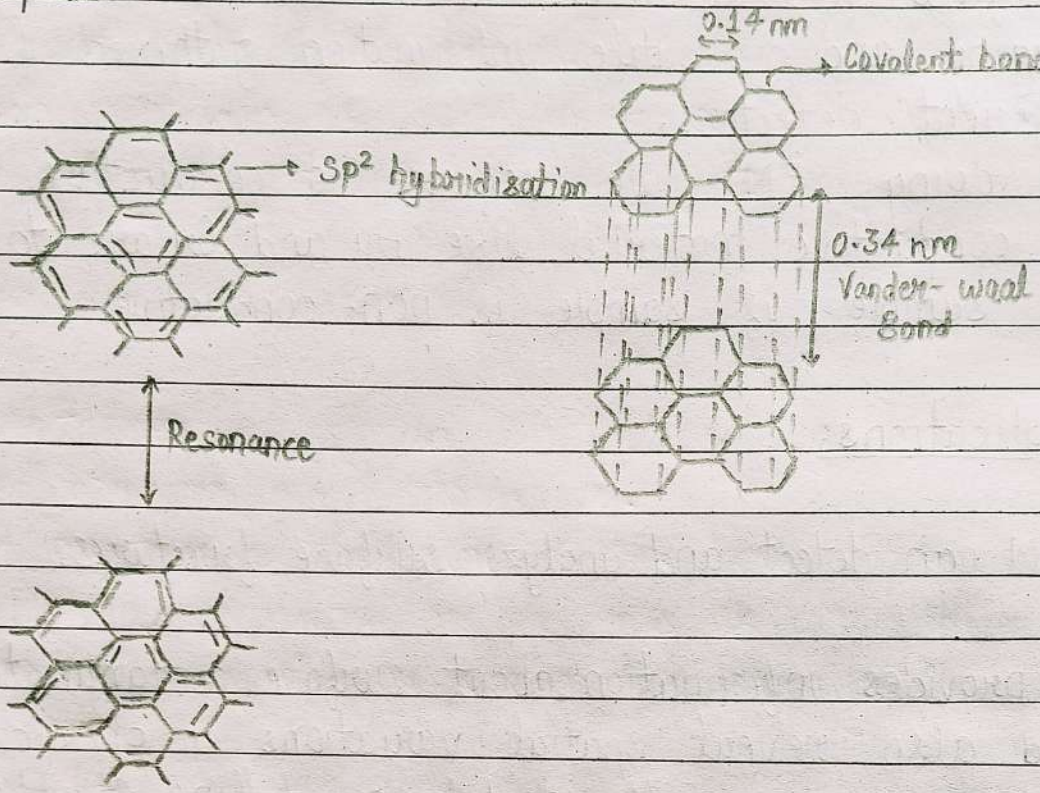
- In a typical SEM, the beam passes through pairs of scanning coils or pairs of deflector plates in the electron column to the final lens, which deflect the beam horizontally and vertically so that it scans in a raster fashion over a area of the sample surface.
- Electronic devices are used to detect and amplify the signals and display them as a distribution map of the intensity of the signal being emitted from the scanned area of the sample on CRT, which later converts into an image.

- It gives detailed three dimensional and topographical imaging and versatile information gathered from different detectors.
- For doing SEM, we require a conductive sample or conductive material like Au and Cr are to be coated on sample if sample is non-conductive.

Applications:

- SEM can detect and analyze surface fractures.
- It provides information about surface contaminations and also reveal spatial variations in chemical compositions as well identify crystalline structures.
- SEMs can be as essential research tool in fields such as life science, biology, gemology, medical & forensic science, metallurgy.
- In addition, SEMs have practical industrial and technological

Graphite:



- Graphite is an allotrope of Carbon.
- Graphite has 2-D solid, consist of C- atoms arranged hexagonally in layers.
- Only 3 valency of each carbon bonded covalently & 4th valency of it is free to move under the influence of current
- In graphite, the sheets are held together by weak Vander waal forces.
- The distance b/w the atoms is 0.142 nm & the distance b/w the adjacent layer is 0.34 nm
- Graphite show sp^2 hyb^n & the bond angle are 120°. with each other.

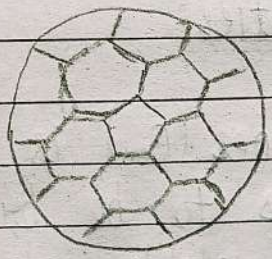
Weak bonding b/w the layers results in the graphite crystal being slippery & soft. It is very good lubricant as the coefficient of friction is low.

Applications:

- Graphite is used in lead pencil.
- Graphite is good lubricant, as coefficient of friction is low
- Graphite is a good thermal & electrical conductor.
- Used as a heating element of electrical furnaces.
- Used as a electrodes for arc welding & in batteries. Also used in mettalurgical crucibles.
- Used in rocket ~~nozzles~~. nozzels.
- In chemical reactor vessels.
- For electrical contacts.
- Brushes & resistors.
- In purification devices
- For high temperature, re-factories & insulation.

Fullerenes:

- Fullerenes are family of carbon allotropes
- It consists of a spherical, ellipsoid or cylindrical arrangement of dozens of C- atom.
- Smallest member of fullerene is C-20 fullerenes & giant fullerene contain C-600 carbon atoms in it.
- Among all the members, C-60 is most important. which is also known as Buky ball or buck-minister fullerenes.



- After the name of its discoverer R. Buckminister fuller, who discovered it in 1985.
- The fullerene molecule consist of 60 carbon atom arranged in pentagons & hexagons to form soccer ball (buky ball) shaped structure.

- It has a truncated icosahedron, which is polygon with C-60 vertices.
- Each pentagon is surrounded by 5 hexagon & no. of two pentagons are adjacent.

Preparation of fullerenes:

- Kraitchner & Huffman synthesised C_{60} by heating of graphite in inert atmosphere.
- A mixture of fullerene 5-15% are obtained which is extracted by solvents of or sublimation can be separated by chromatographically.

Graphite —(Inert atmosphere)→ Mixture of fullerene —(Solvent extraction or sublimation)→ Aluminium hexane separated → Chromatography → C_{60} fullerenes

Properties:

- Fullerenes (C_{60}) is a mustard coloured solid which appears to be brownish black colour.
- It is soluble into aromatic hydrocarbon giving magenta colour.
- It is highly electronegative in nature.
- On exposure to strong UV light, it polymerises, hence it is highly photosensitive in nature.

- On sublimation, it converts magenta crystal.

Applications:

Fullerenes are used -

→ As a photoresistance in certain photolithography process.
→ In the preparation of superconductors when doped with alkali metal.
→ In electronic & multielectronic devices.
→ In optical devices.
→ In batteries as a charge carrier.
→ In the preparation of soft ferromagnet.
→ Carbon nanotube & nanowires are used as a carbon ressins, in golf club, tennis, bicycle frame,
→ Each fibre is times thinner than human fibre.
→ C_{60} is used as a part of catalyst or a substrate.

Polymer are high molecular weight, large molecular which are formed by combination of small units joined together by